

A Grammar of Graphics Approach to Managing Numerical Ties in Parallel Coordinate Plots

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2026-04-01

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1 Introduction and Motivation

Parallel coordinate plots (PCPs) are a powerful technique for investigating patterns across multiple attributes (variables) simultaneously (Inselberg 2009). PCPs assign each dimension of an n -dimensional dataset to a vertical axis, arranged in parallel (Wegman 1990). Observations are drawn as polylines connecting single points along each dimensional axes in sequence a PCPs provide perspective on hard-to-visualize multidimensional data that can be used to identify clusters, outliers, and other facets of a dataset. Simple PCP here, reference in 2nd sentence

When multiple observations share the same value in a given dimension, their polylines perfectly overlap, at a changepoint, creating “visual collisions,” masking information about the joint distribution between variables as well as obscuring the density along a single axis. The treatment of ties is an aspect not generally addressed in the original parallel coordinate plots of Inselberg (1985) and Wegman (1990).

PCPs were canoncially used to show continuous random variables, but generalizations to for categorical data only exacerbated the problem. Categorical parallel axis plot solutions, such as Sankey¹ and Alluvial diagrams, Categorical PCPs (Pilhöfer and Unwin 2013), and Parallel Sets (Kosara, Bendix, and Hauser 2006) plots all developed (largely independently!) as methods for using parallel

¹Sankey diagrams do not always have parallel axes, but they are visually similar and frequently use parallel axis constructions.

axes to represent categorical, bivariate frequency information. Variations on these plots, such as common angle plots (Hofmann and Vendettuoli 2013), address perceptual distortions that arise when line widths are evaluated along the vertical rather than the orthogonal direction, offering an alternative encoding that more faithfully communicates categorical association strengths.

Hammock plots (Schonlau 2003), another type of PCP that accommodates categorical and continuous variables, shows multiple bivariate relationships using parallelograms drawn between axes, obviating the need for tie-breaking and preserving the density information. It is the parallelogram construction that creates the bivariate limitation. Because only bivariate relationships are shown, it is not possible to mentally reconstruct the full joint distribution using a hammock plot.

You need a hammock plot here to reference -- one that shows a continuous variable

The development of generalized PCPs, as implemented in `ggpcp` (VanderPlas et al. 2023)) represents a major step forward in representing the full joint distribution of the data while showing individual observations. One innovation in the `ggpcp` package is the handling of ties in continuous variables: ties are broken and the observations are spread out within the box representing the marginal frequency of each value of the categorical variable. Sorting methods are implemented to distribute the tied values in a way that reduces the overall complexity of the plot, as random line-crossings can make PCPs difficult to read and interpret.

categorical,

This project aims to extend the treatment of categorical variables in generalized PCPs, implementing a solution for the treatment of continuous variables which has the following properties:

- marginal density information (approximately) faithfully represented on the vertical axis
- individual lines can be visually distinguished, at least for data where the number of observations does not induce overplotting.

used to handle categorical variables and implement a similar method for continuous ties

An immediate consequence of the second objective is that the viewer can, at least in theory, reconstruct the full joint distribution between variables from the plot. We will extend the method used in `ggpcp` to represent categorical data to continuous variables, modifying it to maintain the approximate scaling of the continuous variable, with slight local distortions when tied values occur. In addition, we will develop a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values, while preserving the visual representation of singular values on the parallel axis.

This combination of the advantage of Hammock plots – maintaining the marginal density information – and generalized PCPs, which preserve individual observations and provide the ability to reconstruct the full joint distribution, will enhance GPCPs. Real-world data sets frequently have both categorical and numerical data, and duplicated values in either type of variable. Our extension to GPCPs will facilitate the visual representation of these data sets.

1.1 Related Work

The parallel coordinate plot literature has developed several threads directly relevant to the tie-breaking problem addressed here.

Overplotting and density. Early responses to visual overplotting in PCPs focused on transparency and kernel density estimation rather than positional adjustment. Alpha-blending, as used by Miller and Wegman (1990), allows the density of overlapping lines to become visible through

saturation, but sacrifices the ability to trace individual observations. More recent approaches include density-based rendering (Heinrich and Weiskopf 2013) and edge bundling (McDonnell and Mueller 2008), both of which trade individual traceability for aggregate clarity.

Categorical extensions. The categorical parallel axis plot literature developed largely independently of classical PCPs. Parallel Sets (Kosara, Bendix, and Hauser 2006) replaced individual polylines with ribbons whose widths encode bivariate frequencies, enabling unambiguous density reading at the cost of individual tracing. Alluvial diagrams (Brunson 2020) and Sankey diagrams adopt a similar ribbon-based representation. Common angle plots (Hofmann and Vendettuoli 2013) addressed a further perceptual distortion specific to variable-width ribbons: the Müller-Lyer family of line-width illusions causes ribbon widths evaluated along the vertical to appear systematically different from their orthogonal widths. By constraining ribbons to a constant angle, common angle plots recover perceptually accurate frequency judgments. Hammock plots (Schonlau 2003; Schonlau and Yang 2024) occupy a different position: they accommodate both categorical and continuous variables via constant-width parallelograms, but restrict inference to bivariate relationships between adjacent axes.

Axis and category ordering. A well-studied property of PCPs is that axis order strongly affects both the number of line crossings and the interpretability of patterns (VanderPlas et al. 2023). Minimizing line crossings through reordering is NP-hard in general (Martí and Laguna 2003), but heuristics based on correlation structure and sum of ranking differences provide tractable approximations (Ipkovich, Héberger, and Abonyi 2021). Within a fixed axis ordering, the order of factor levels and the plotting order of individual observations interact with tie-breaking: factor levels ordered to minimize crossings produce fewer extraneous crossings for any given tie-resolution strategy, while the `overplot` parameter in `ggpcp` controls which observations are rendered on top when lines overlap (VanderPlas et al. 2023). These considerations interact directly with the hierarchical sorting approach developed in this proposal and are discussed further in Section 2.2.

Computational quality metrics. Dennig et al. (2021) formalized a suite of screen-space quality metrics for Parallel Sets—including ribbon overlap, crossing angle, orthogonality, and ribbon width variance—collectively termed ParSetgnostics. These metrics provide an empirical basis for comparing and optimizing parallel axis layouts without requiring user studies, and form the basis for the computational benchmarking component of our evaluation plan (Section 4.3).

1.2 Categorical Variables and Ties in `ggpcp`

The `ggpcp` package currently addresses categorical ties through a combination of sorting and tie-breaking algorithms. The package implements hierarchical sorting through the `pcp_arrange(data, method, space)` function, which orders observations based on a hierarchical application of variable values. The `method` parameter determines the sequence in which variables are considered when resolving ties in the arrangement. The `space` parameter specifies the proportion of the y-axis dedicated to empty space between levels of categorical variables (Figure 1).

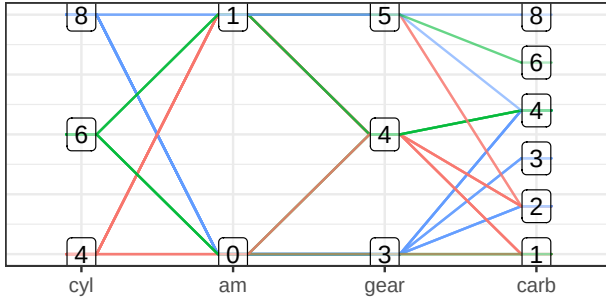
```
# 1. Modular Data Pipeline (The correct Step 1, 2, 3)
# Note: 'pcp_data' is used here as a variable name for the transformed object.
pcp_data <- mtcars %>%
  mutate(across(c(cyl, am, gear, carb), as.factor)) %>%
  pcp_select(cyl, am, gear, carb) %>%      # Step 1: Reshape data
  pcp_scale(method = "uniminmax") %>%     # Step 2: Scale axes
```

```
pcp_arrange(method="from-left")
```

```
# Step 3: Break Ties,  
#         sorting from left
```

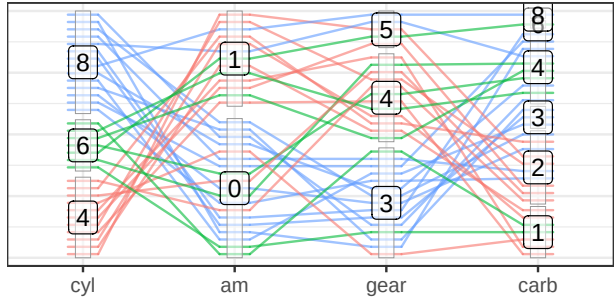
Standard PCP

No Tie Breaking



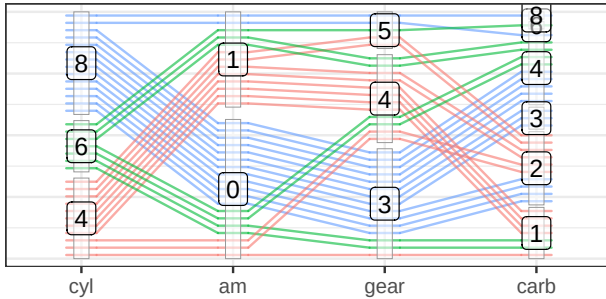
GPCP: Tie Breaking

No sorting



GPCP: Cat. Ties + Sorting

Sort from left



GPCP: Cat. Ties + Sorting

Sort from right

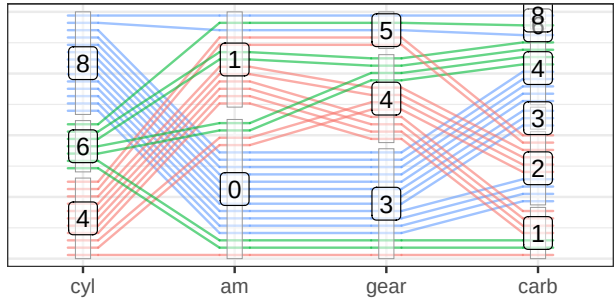


Figure 1: Comparison of tie-breaking methods for categorical variables in parallel coordinate plots using mtcars data. Standard PCP (top-left) shows overlapping lines without tie-breaking. GPCP with tie-breaking but no sorting (top-right) spreads observations evenly within categories. GPCP with sorting from left (bottom-left) and from right (bottom-right) apply hierarchical sorting to minimize line crossings, with light gray boxes indicating category groupings.

The ability to follow individual observations is key to being able to reconstruct the full joint distribution across parallel axes. `ggpcp` creates equally spaced points within each category that span the portion of the vertical axis dedicated to the category (preserving the marginal frequency information, which is optionally emphasized with `geom_pcp_boxes`). The top left plot in Figure 1 shows the PCP without this categorical tie-breaking approach. However, `ggpcp` goes one step further by using hierarchical sorting (from the left, right, or both) to minimize unnecessary line crossings induced by the treatment of tied categorical variables. The absence of the hierarchical sorting approach is emphasized in the top right plot in Figure 1, while the bottom left and right plots show sorting from the left and right respectively. This hierarchical sorting approach serves as a form of “external cognition,” reducing the cognitive load required to “untangle” and group crossed lines. The sorting also ensures that only necessary crossings² occur, minimizing visual clutter.

It is worthwhile to examine how a viewer would follow a line across the plot in detail, using a generalized PCP with both numeric and categorical variables to illustrate the challenges introduced

²A crossing is necessary if it is induced by the order of x-axis variables and scaled values, rather than by the treatment of categorical variables.

when numerical ties are present.



Figure 2: The parallel axes form the structural framework of the visualization.



Figure 3: The starting point is identified on the first axis with its corresponding value.

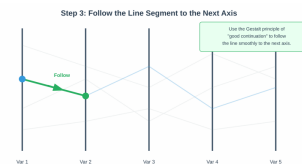


Figure 4: Following the line segment uses the Gestalt principle of good continuation.

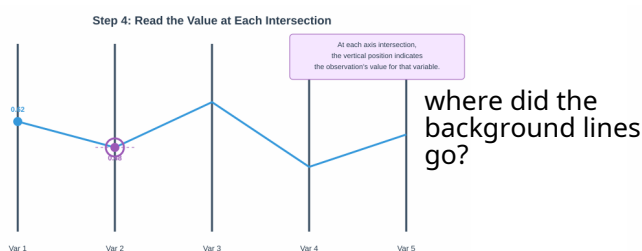


Figure 5: Values are read as approximate scaled value relative to the total range.

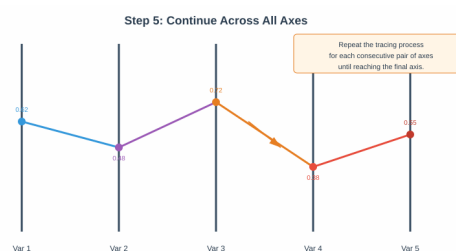


Figure 6: The complete traced path reveals the observation's values across all variables.

- Step 1: Identify the Parallel Axes (Figure 2).
Begin by identifying each vertical axis in the plot. Each axis represents one variable from the dataset. The axes are typically arranged from left to right, and the order may be determined by the data analyst to highlight specific relationships or minimize visual clutter.
- Step 2: Locate the Starting Point (Figure 3).
Find the observation of interest on the leftmost axis. The vertical position indicates the scaled value of that observation for the first variable. If you are examining a highlighted or color-coded observation, look for its distinctive marker at this starting position.
- Step 3: Follow the Line Segment (Figure 4).
Trace the line segment from the starting point to its intersection with the next axis. The human visual system naturally follows smooth, continuous paths due to the Gestalt principle of good continuation, which allows viewers to perceive connected line segments as a single, piecewise continuous line, making it easier to track observations across multiple parallel axes.
- Step 4: Read Values at Intersections (Figure 5).
At each axis intersection, the vertical position of the line indicates the observation's value for that variable. Read these values to understand how the observation changes (typically, relative to the other observations) across different dimensions of the data. The slope of line segments between axes provides information about the relationship between consecutive variables for that specific observation.
- Step 5: Continue Across All Axes (Figure 6).
Repeat the tracing process for each consecutive pair of axes until reaching the rightmost axis.

By following the complete path, you obtain a comprehensive view of how that particular observation behaves across all measured variables. The vertical positions along each axis represent scaled values that can be interpreted as quantiles when the data are appropriately transformed. This enables identification of unique characteristics, cluster membership, or outlier status.

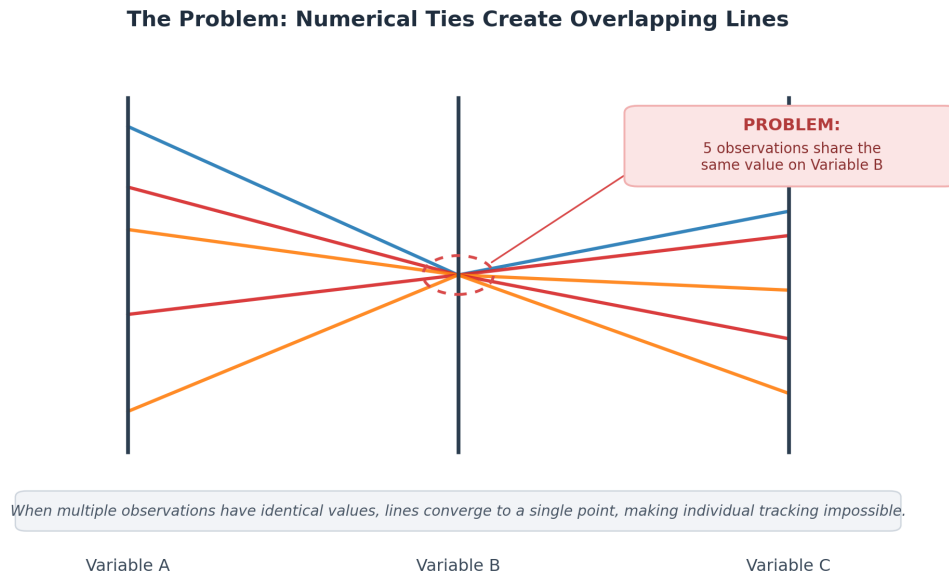


Figure 7: Numerical ties cause lines to converge at a single point. Line identity at the tie can only be recovered when lines arrive with sufficiently distinct colors; here the blue line is distinguishable by opacity, but the cyan and red lines are not cleanly separable, and the values each observation takes on Variables B and C are not uniquely determined from the plot alone. A revised version of this figure will use 2–3 categorically distinct colors to make this failure mode explicit.

- Step 6: Ties in Numbers — A Fork in the Road (Figure 7).

What was once a single point now becomes the starting point for two separate line segments that go in different directions. The Gestalt principle of good continuation doesn't work anymore when two or more lines come from the same point on an axis. In this configuration, color is the only channel that can provide separate line identity across the tie — and even then, only when the palette is carefully chosen. The visual continuity that made tracing easy in earlier steps breaks down exactly where it is needed most. The observer cannot tell if the original observation follows the upper branch or the lower branch of the fork unless there are more visual cues, like color coding, clear labels, or a systematic tie-breaking arrangement. This lack of clarity is not just an aesthetic problem; it also makes it harder to do the main analytical job of the parallel coordinate plot, which is to show patterns that would be hard to see in tabular data by following individual observations through high-dimensional space.

Within the box, use horizontal segments,
diverging on either side of the box (gestalt)

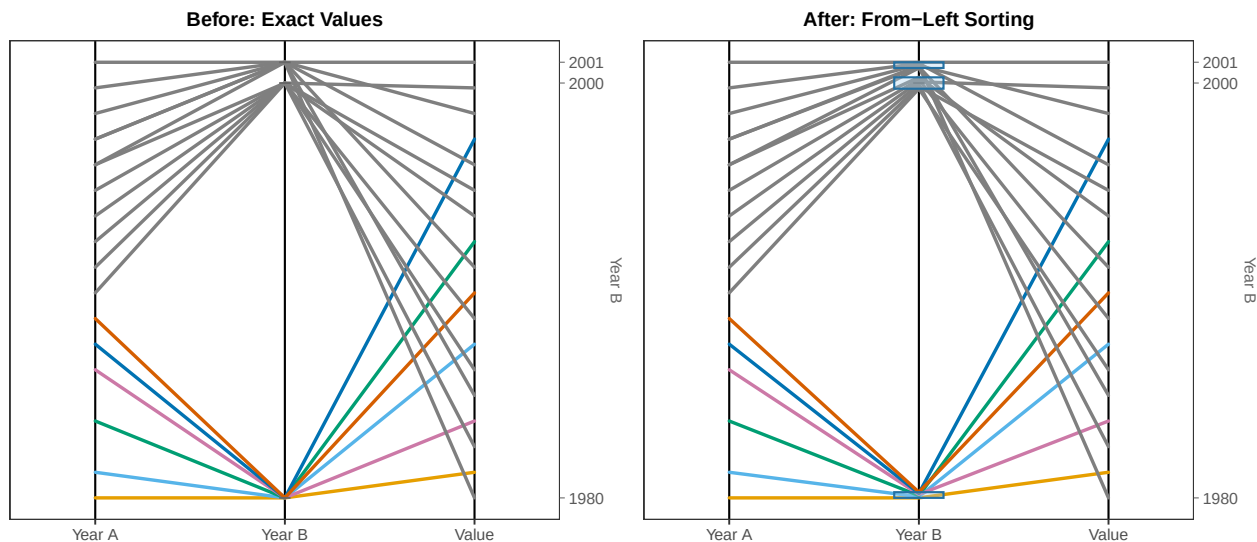


Figure 8: Comparison of a standard PCP with overlapping ties (left) and the proposed tie-breaking algorithm with **from-left** hierarchical sorting (right). Tied observations on Year B are spread within a resolution-based interval (blue boxes) and ordered by their Year A values, eliminating within-band crossings. A visible gap is preserved between the tie bands for 2000 and 2001: separation between bands is essential for the Gestalt grouping argument to hold, since adjacent bands that touch would visually merge and the reader would lose the ability to distinguish distinct values. The band width is therefore set strictly below the global axis resolution δ_j . Remaining crossings are necessary crossings induced by the data structure rather than artifacts of tie-breaking.

1.3 Addressing Numerical Ties

Two implementation details in Figure 8 deserve explicit comment because both are load-bearing for the Gestalt argument that motivates the spreading. First, each tie band must be strictly narrower than the gap to its nearest neighbor: if the band around $v = 2000$ extended up to meet the band around $v = 2001$, the two values would visually fuse, defeating the separation the spread was meant to make visible. The band width in the figure is therefore set to a fraction of the axis resolution δ_j , leaving a deliberate gap between adjacent bands. Second, the reader must be able to see that gap — not merely that it exists in the coordinate system. This is a Gestalt-of-proximity requirement: if ties at adjacent values are rendered without visible separation, the grouping cue that “observations within a band share a value” competes with the unintended cue that “bands adjacent to one another belong to the same group.” The mathematical formalization of this separation requirement appears in Step 3 of the algorithm (Section 3.3.3).

The **ggpcp** package implements a sophisticated tie-breaking algorithm for categorical variables that maintains individual observation traceability. The approach spaces observations evenly within categorical levels:

“All observations are spaced out evenly. This results in a natural visualization of the marginal frequencies along each axis (additionally enhanced by the light gray boxes grouping observations in the same category) that is not as prominent in the previous three panels. The ordering of the observations within the level is such that a minimal number of line crossings occurs between the axes.” (p. 11)

Even spacing alone, however, is not sufficient: without a principled ordering within each tie band, lines departing toward different positions on the neighboring axis will cross inside the band, reproducing the very confusion the spreading was meant to resolve. The hierarchical sorting approach that resolves this is described in detail in Section 2.2; the spacing formula used for categorical levels is:

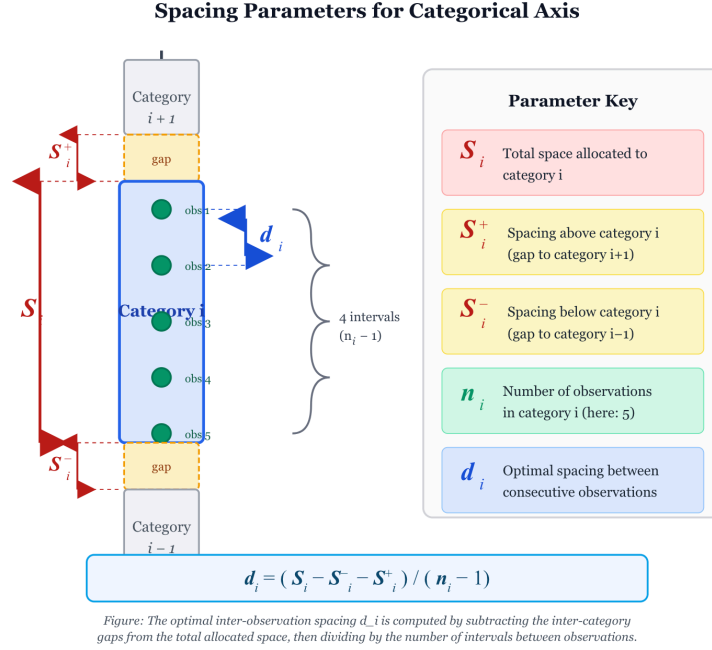


Figure 9: Spacing parameters for Categorical Axis

$$d_i = \frac{S_i - S_i^- - S_i^+}{n_i - 1}$$

where:

- S_i is the total space allocated to category i
- S_i^- is the spacing below category i
- S_i^+ is the spacing above category i
- n_i is the number of observations in category i
- d_i is the optimal spacing distance between consecutive observations

A key visual difference emerges when connecting categorical to numerical variables. Schonlau and Yang (2024) observe:

“When many observations have the same value for a categorical and an adjacent numerical variable, the corresponding area looks like a triangle... Notice the lines/boxes between the variables hospitalizations and comorbidities in the GPCP (Figure 13) and hammock plots (Figure 2). Most of the observations are in the boxes leading from hospitalizations=0 to either comorbidities=0 or comorbidities=1. This is far more obvious in the hammock plot than in the GPCP plot.” (p. 19)

Whether the transition takes a triangular or rectangular form is perhaps less important than what it communicates about the data itself. The more interesting question is whether the visualization preserves density information while still allowing viewers to follow individual observations. When observations flow from a categorical box toward numerical values, tie-breaking on the numerical axis shapes how clearly we can perceive the underlying distribution.

The `ggpcp` package tackles this challenge through its tie resolution parameters. The `space` parameter controls vertical spacing between tied values, and the `method` parameter governs how those values are ordered. Used well, these parameters spread observations along the numerical axis much like a rug plot would, giving readers a sense of density while keeping individual paths traceable. This represents a different set of trade-offs than hammock plots, which use constant-width parallelograms to show marginal densities directly. Rather than foregrounding aggregate density, `ggpcp` keeps individual observation lines intact and relies on thoughtful spacing to convey distributional structure. The result is a visualization that supports both seeing the forest and examining particular trees.

The preceding review of categorical tie-breaking, together with the related work in Section 1.1, sets the stage for a formal specification of the research questions this proposal addresses.

1.4 Research Questions

The preceding review motivates three formal research questions that structure the remainder of this proposal.

RQ1 (Algorithm correctness). Does the `tie_spread` algorithm distribute tied observations within resolution-based intervals in a way that (a) preserves marginal frequency information on the vertical axis and (b) eliminates within-band line crossings through hierarchical sorting?

RQ2 (Frequency perception). Do viewers make more accurate frequency judgments—for magnitude estimation, ordinal comparison, and ratio tasks—when examining a GPCP with numerical tie-breaking than when examining an equivalent hammock plot?

RQ3 (Individual traceability). Does numerical tie-breaking in GPCPs improve viewers’ ability to follow a single observation across all axes, relative to a standard PCP without tie-breaking, without sacrificing the density information conveyed by the marginal axis representation?

RQ1 is addressed computationally through the algorithm development and visual inspection phases (Phase 1 and 3, Section 4.2). RQ2 and RQ3 are addressed through the perceptual validation study (Phase 2). We pre-specify directional hypotheses for RQ2 and RQ3: we anticipate that GPCP with tie-breaking will produce lower absolute percentage error on magnitude estimation tasks (RQ2) and higher tracing accuracy (RQ3) than the respective baselines, with effect sizes consistent with a medium Cohen’s d based on prior graphical perception literature.

2 An Approach to Numerical Ties

Currently, numerical variables disrupt some of the best innovations in the GPCP approach by making it impossible to track an observation across the plot and by obscuring marginal density information through overplotting. The remaining two chapters of the dissertation will address the handling of ties on numerical axes and assess the utility of visual cues that can be paired with the

tie breaking method to indicate that values are approximately accurate spatially and equivalent numerically.

The `ggpcp` package does not currently provide a mechanism for resolving these tied values. We propose `tie_spread`, an algorithm that distributes observations with identical values along the vertical axis, transforming overlapping lines into a visually resolvable spread. Consistent with tidyverse conventions and the tidyselect grammar, the implementation allows users to specify tie-breaking behavior selectively across axes—applying spreading to some variables while preserving exact positioning on others. Sensible defaults balance visual clarity against faithful representation of the underlying data, ensuring that users can immediately benefit from improved legibility without extensive parameter tuning. The following section details the design and implementation of this approach.

2.1 The Problem: Overlapping Lines

When multiple observations share the same value on a numerical axis, their polylines converge to a single point on that axis, producing perfectly overlapping line segments on either side. This convergence creates two compounding problems: individual observation traceability is lost, because the viewer cannot distinguish which incoming line corresponds to which outgoing path, and marginal density information is obscured, because a cluster of ten tied observations is visually indistinguishable from a single one. The problem is particularly acute for integer-valued or coarsely-measured variables, where ties are structurally common rather than incidental (Figure 7).

The `tie_spread` algorithm resolves tied observations by distributing them evenly within a proportionally allocated vertical band on the axis. Each unique value v_i receives space proportional to its frequency:

letter representing space, please

$$S_i = \frac{n_i}{\sum_k n_k} (1 - \text{space} \cdot (m - 1))$$

where n_i is the number of observations tied at v_i , m is the number of distinct values on the axis, and `space` is the proportion of the axis reserved for gaps between value bands. Observations are then spread evenly from the lower to the upper boundary of their allocated band, with small buffer margins at the edges. This directly mirrors the categorical tie-breaking approach already implemented in `ggpcp`, extending it to numerical variables and thereby creating a unified visual grammar across both variable types (Figure 10). Values with more tied observations occupy a taller band, implicitly encoding marginal frequency in the vertical extent of each region—an analog to the light gray boxes used for categorical levels.

2.2 Hierarchical Sorting for Minimal Crossings

Spreading tied values evenly across a resolution-based interval is a necessary first step, but it is not, by itself, sufficient to eliminate visual knots. When the order of assignment within the tie band is arbitrary, lines departing toward different positions on the neighboring axis will cross inside the band, reproducing the very confusion the spreading was meant to resolve. The `ggpcp` package resolves this through hierarchical sorting that mirrors the approach already used for categorical variables.

horizontal segments, and show more angles outside of the boxes via endpoints.

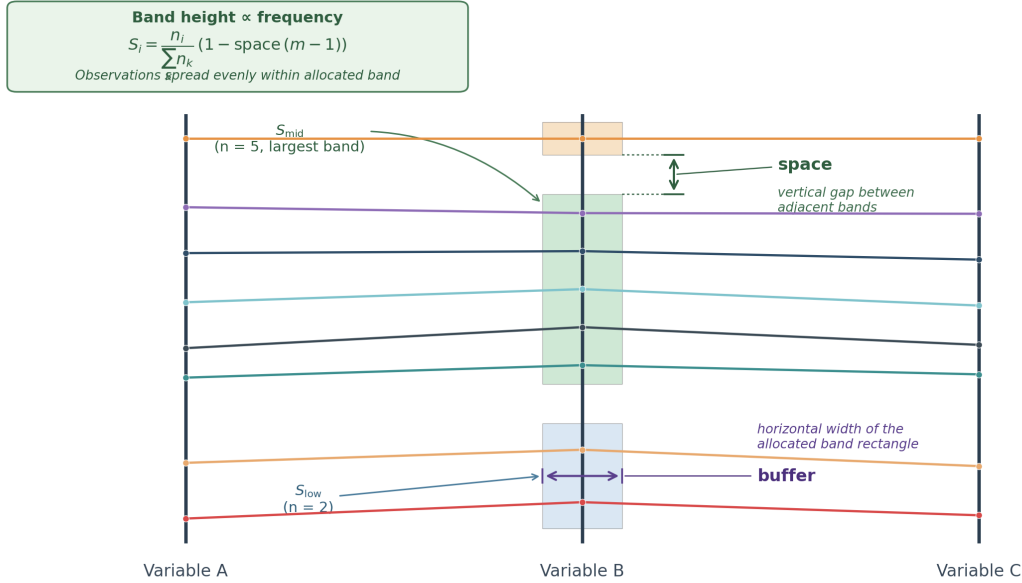


Figure 10: Proportional band allocation for numerical ties. Each unique value on the numerical axis receives a vertical band whose height is proportional to its frequency ($S_i \propto n_i$), and observations are spread evenly from the lower to the upper boundary of their allocated band. Three visual elements are marked in the figure: the **band height** (S_i), which encodes marginal frequency; the **band separation** (the **space** parameter), the unencoded gap between adjacent bands that preserves the Gestalt cue distinguishing distinct values; and the **edge buffer**, a small margin inside each band that prevents observations from landing exactly on the band boundary. The largest band in this example contains five tied observations ($n = 5$); the smallest contains two ($n = 2$). This design mirrors the categorical tie-breaking approach in `ggpcp`, ensuring consistent visual grammar across variable types.

For the **from-right** processing mode — the mode directly applicable to Figure 8 — each tied observation $i \in T_j(v)$ is ranked by its adjusted position on the right neighbor axis, $\kappa_i = \tilde{y}_{i,A_j}$ (see Equation 8). Positions within the tie band are then assigned in the same rank order, from lowest to highest. This monotone rank-matching guarantees that no two lines cross within the tie band: the observation that arrives lowest on the right axis is placed lowest in the band, the next-lowest observation is placed next, and so on. Any crossings that remain after sorting are *necessary* crossings — those induced by the data structure itself rather than by the mechanics of tie-breaking.

By ordering observations within a tie group based on their values on adjacent axes, the package leverages the Gestalt principle of common fate. This strategy ensures that observations with similar trajectories are positioned in close proximity, creating cohesive visual bands. These bands facilitate the perception of distinct groups as they move together through high-dimensional space, reducing visual noise and highlighting underlying patterns.

Interaction with axis ordering. The effectiveness of hierarchical sorting is not independent of axis order. VanderPlas et al. (2023) note that the order of factor levels is an important determinant of residual line crossings: some crossings are attributable to the data structure itself and cannot be eliminated by within-category sorting, while others arise from an unfavorable factor ordering and can be removed by reordering levels. The **ggpcp** package allows users to reorder factor levels via **mutate** before variable selection, and to reverse axes to reduce negatively correlated crossings. Our tie-breaking extension inherits this dependency: for numerical axes, the minimum number of necessary crossings is a function of both the scaled data values and the sequence in which axes are processed. Users should therefore consider axis ordering as a precondition for optimal tie-breaking, particularly when integer-valued variables with many ties are adjacent to continuous variables with high variance.

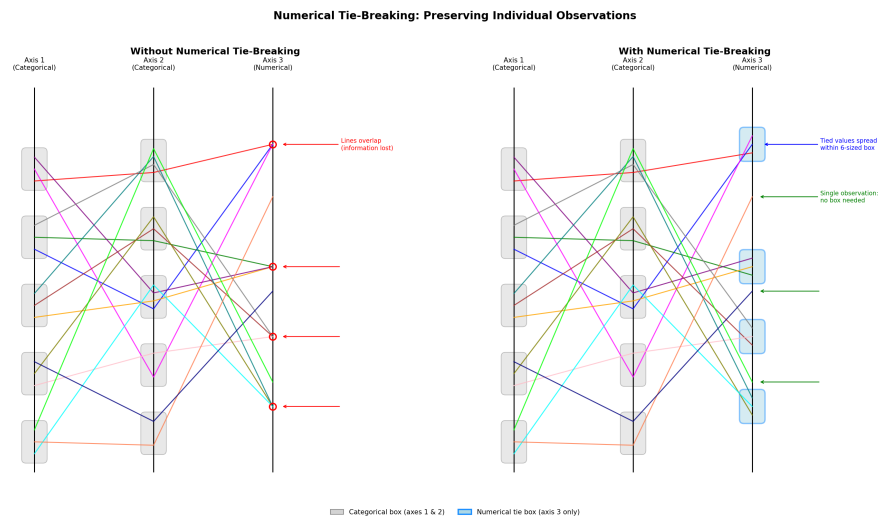


Figure 11: Comparison of parallel coordinate plots without (left) and with (right) numerical tie-breaking using **from-right** hierarchical sorting. When observations share the same numerical value, their lines overlap completely (left), hiding how many cases exist at that value. Spreading tied values within resolution-based boxes (blue) and sorting by the right neighbor axis (right) makes each observation visible, eliminates crossings within the tie band, and signals that positions within the box are artifacts of tie-breaking rather than meaningful data variation.

2.3 Gestalt Principles and Line Continuity

The principle of good continuation states that the human visual system preferentially perceives smooth, continuous contours over interpretations requiring abrupt changes in direction. This principle directly impacts the effectiveness of parallel coordinate plots for tracing individual observations.

Equispaced lines maintain continuity by representing each observation as a continuous polyline extending from the leftmost to the rightmost axis. When observations share identical numerical values (ties), the equispacing mechanism distributes lines within each category so that they remain visually distinct without introducing discontinuities.

Constant-width boxes represent observations as aggregated area segments rather than individual lines. This encoding communicates aggregate quantities effectively: the area of each segment is proportional to the number of observations sharing that combination of values across adjacent axes, making the joint frequencies of category pairs immediately visible. In contrast, equispaced lines preserve individual observations as distinct visual elements. When lines are continuous and uninterrupted, viewers can leverage preattentive processes such as the Gestalt principle of good continuation to trace a single case across all axes (Healey and Enns 2012). The two approaches thus serve distinct purposes: lines support individual-level tracing through low-level perceptual grouping, while boxes support aggregate-level frequency comparison. The conditions under which each is optimal are taken up in the Conclusion.

2.4 Visual Cue Design

The introduction committed to developing “a consistent visual representation to provide visual cues which indicate the presence of a local distortion due to tied values.” This section specifies the design rationale for those cues, building on the perceptual principles outlined in the preceding section.

When tied values on a numerical axis are spread within a resolution-based interval, the positions of observations within that interval are artifacts of tie-breaking rather than reflections of meaningful data variation. A viewer who is unaware of this may misread the spread as genuine within-group variance. The visual cue must therefore accomplish two things simultaneously: it must make the presence of a tie region legible without disrupting the tracing of individual lines through the region.

The design adopted here uses a light gray bounding box, anchored at the lower and upper bounds of the resolution interval $[L(v), U(v)]$, as illustrated in Figure 13 and Figure 14. This choice follows directly from the categorical tie-breaking convention already implemented in `ggpcp`, where gray boxes group observations within each categorical level. Extending the same visual element to numerical tie regions creates a unified grammar: a gray box in any axis region signals “these positions are spread for legibility, not because the underlying values differ.” This consistency reduces the cognitive overhead required to interpret mixed categorical-numerical plots.

Three alternative cue designs were considered and set aside. A color change on tied lines would require an additional channel already commonly used for group membership. A texture fill would introduce ambiguity on screen at small box heights. A dashed axis segment at the tied value would mark the location but not the extent of the spread, making it difficult to recover the original value. The gray box avoids all three problems: it marks both the location (the center of the box

corresponds to the tied value) and the extent (the box height reflects δ_j , the global axis resolution), and it does not consume an encoding channel needed elsewhere.

The `get_tie_box()` function implements this detection and rendering pipeline in three steps: identifying values that appear more than once on a given axis, grouping contiguous spreads at numerical precision, and generating bounding box coordinates for `geom_rect` rendering. An optional label argument displays the original tied value at the center of each box, which is particularly informative for integer-valued variables where many values may be tied at round numbers.

3 Mathematical Framework

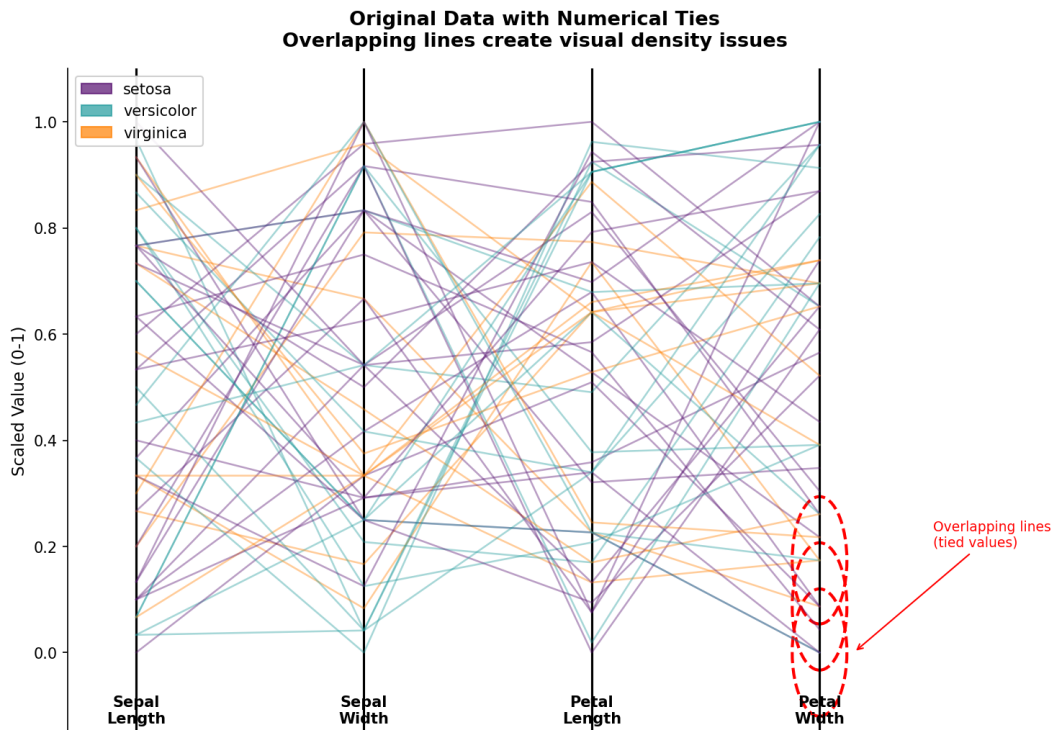


Figure 12: The tie problem in parallel coordinate plots. When observations share values, lines overlap and create visual collisions that mask the underlying data distribution.

3.1 Notation

Let $\mathcal{D}$ be a dataset with n observations and p variables displayed on parallel axes $X_1, X_2, \dots, X_p$.

Table 1: Core notation

Symbol	Definition
x_{ij}	Raw value of observation i on variable j
y_{ij}	Scaled value (normalized to $[0, 1]$)
$\tilde{y}_{ij}$	Adjusted value after tie-breaking

Symbol	Definition
$T_j(v)$	Set of observations tied at value v on axis j
k	Number of tied observations: $k = T_j(v) $
δ_j	Resolution of axis j : minimum distance between distinct values

3.2 Scaling

Variables are normalized to $[0, 1]$ using min-max scaling:

$$y_{ij} = \frac{x_{ij} - \min(X_j)}{\max(X_j) - \min(X_j)} \quad (1)$$

3.3 The Tie-Breaking Algorithm

3.3.1 Step 1: Compute Axis Resolution

To preserve density information and maintain consistent spacing, we first compute the global resolution of axis j —the smallest difference between any two distinct values:

$$\delta_j = \min\{|y_1 - y_2| : y_1 \neq y_2, y_1, y_2 \in Y_j\} \quad (2)$$

where Y_j is the set of unique values on axis j .

Using a consistent jitter range across all values ensures that the visual representation preserves both the density of observations at each value and the gaps between distinct values. An approach that computed jitter ranges individually for each tied value would risk collapsing visually meaningful gaps, destroying distance information in the display.

3.3.2 Step 2: Identify Ties

A value v on axis j is a tie if multiple observations share it:

$$T_j(v) = \{i : y_{ij} = v\}, \quad |T_j(v)| > 1 \quad (3)$$

3.3.3 Step 3: Compute Available Space

Tied observations at value v are spread within an interval defined by the axis resolution:

$$L(v) = v - \frac{\delta_j}{2}, \quad U(v) = v + \frac{\delta_j}{2} \quad (4)$$

If v is at the axis boundary, apply a buffer β (default 0.05):

- $L(v) = \max(0, v - \delta_j/2)$ if $v - \delta_j/2 < 0$
- $U(v) = \min(1, v + \delta_j/2)$ if $v + \delta_j/2 > 1$

The available space is:

$$S(v) = U(v) - L(v) \quad (5)$$

Separation constraint. Defining the band width as exactly δ_j would, for evenly-spaced distinct values, produce bands that share endpoints with their neighbors and therefore render as a single continuous column on the axis. This collapses the visual separation between distinct values and defeats the Gestalt grouping cue that the spread was intended to establish (see Figure 8). We therefore introduce a separation parameter $\sigma \in (0, 1]$ (default $\sigma = 0.9$), so that the effective band width is $\sigma \cdot \delta_j$ and the bounds become:

$$L(v) = v - \frac{\sigma \cdot \delta_j}{2}, \quad U(v) = v + \frac{\sigma \cdot \delta_j}{2} \quad (6)$$

The residual fraction $(1 - \sigma) \cdot \delta_j$ is left as an unencoded gap between adjacent tie bands, ensuring that the reader can perceive distinct values as distinct regions rather than as a continuous strip.

3.3.4 Step 4: Compute Optimal Spacing

The core principle is identical for categorical and numerical ties:

$$\Delta = \frac{S(v)}{k - 1} \quad (7)$$

This distributes k observations evenly across the available space $S(v)$.

3.3.5 Step 5: Order Observations Hierarchically

To minimize line crossings, tied observations are sorted by their positions on the adjacent axis before spread positions are assigned. For axis j , let A_j be the adjacent axis — the left neighbor under **from-left** processing, or the right neighbor under **from-right** processing.

For each observation $i \in T_j(v)$, the sorting key is:

$$\kappa_i = \tilde{y}_{i, A_j} \quad (8)$$

Observations are ordered so that $\kappa_{i_1} \leq \kappa_{i_2} \leq \dots \leq \kappa_{i_k}$.

This ordering is the step that resolves the visual “knots” illustrated in Figure 8. Spreading alone distributes tied points across the interval $[L(v), U(v)]$, but without a consistent rank-matching to the neighbor axis, lines departing the band will cross inside it. Sorting by κ_i before assigning positions in Equation 9 creates a monotone mapping: the observation placed at rank m in the tie band is exactly the observation ranked m -th on the neighbor axis. Under **from-right** processing, this means the lowest position in the band connects to the lowest position on the right axis,

eliminating within-band crossings entirely. Any crossings that remain after this step are necessary crossings induced by the data structure itself.

Key insight: This requires *sequential* axis processing — the adjusted positions from previously processed axes inform the ordering on the current axis.

3.3.6 Step 6: Assign Positions

Given ordered observations $(i_1, i_2, \dots, i_k)$:

$$\tilde{y}_{i_m, j} = L(v) + (m - 1) \cdot \Delta, \quad m = 1, 2, \dots, k \quad (9)$$

3.3.7 Position Functions

Two variants of position functions distribute observations within the available space:

The edge-inclusive version places observations at:

$$\text{positions} = L(v) + \frac{i - 1}{k - 1} \cdot S(v), \quad i = 1, \dots, k$$

3.4 Unified Framework

The algorithm treats numerical ties identically to categorical levels:

Table 2: Unified spacing principle

Variable Type	Available Space	Spacing
Categorical	Box height h_ℓ for level ℓ	$\Delta = h_\ell / (n_\ell - 1)$
Numerical	Resolution-based interval $S(v) = \delta_j$	$\Delta = S(v) / (k - 1)$

Both apply the same formula: **spacing** = **available_space** / (**n** − 1).

3.5 Visual Results

3.5.1 Uniform Spacing Applied

After applying the tie-breaking algorithm, observations that previously overlapped are now visually distinguishable:

3.5.1.1 Annotated Tie Regions

Tie regions can be annotated with boxes showing the original value and the spread region:

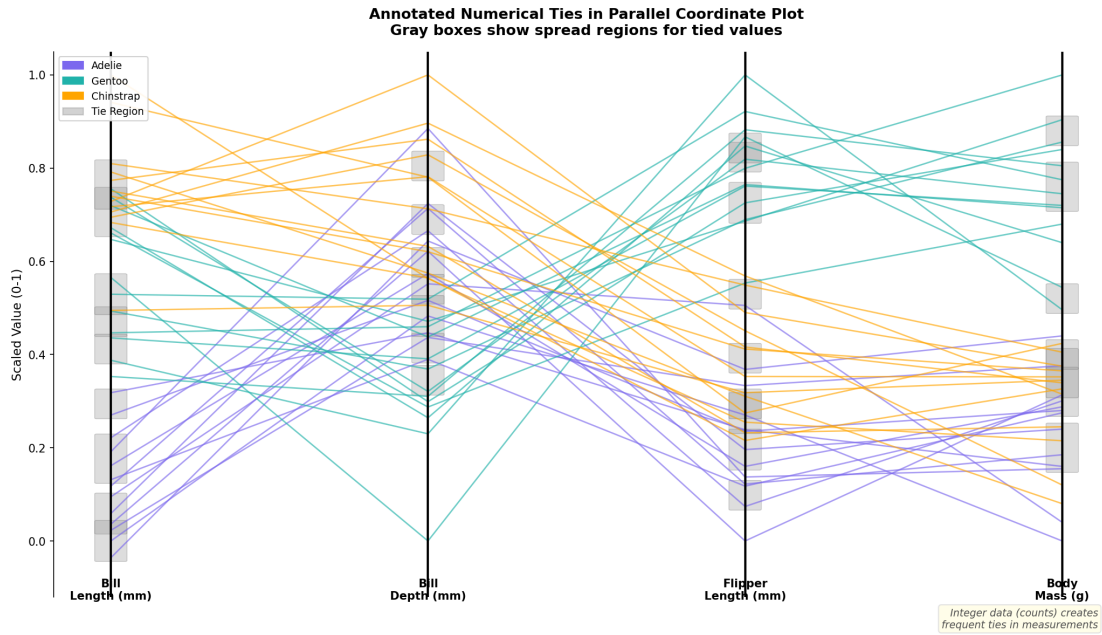


Figure 13: Parallel coordinate plot after applying uniform spacing. Gray boxes indicate regions where tied values have been spread, making individual observations traceable.

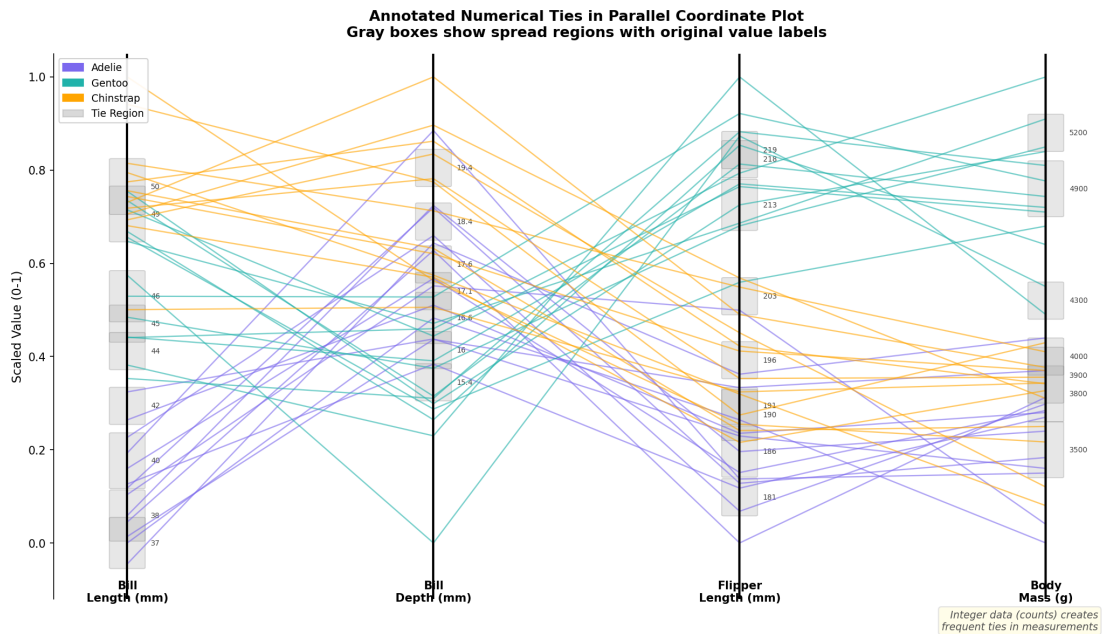


Figure 14: Annotated tie regions with labels. Gray boxes highlight where ties occurred, with labels showing the original tied value. This is particularly useful for integer data where ties are frequent.

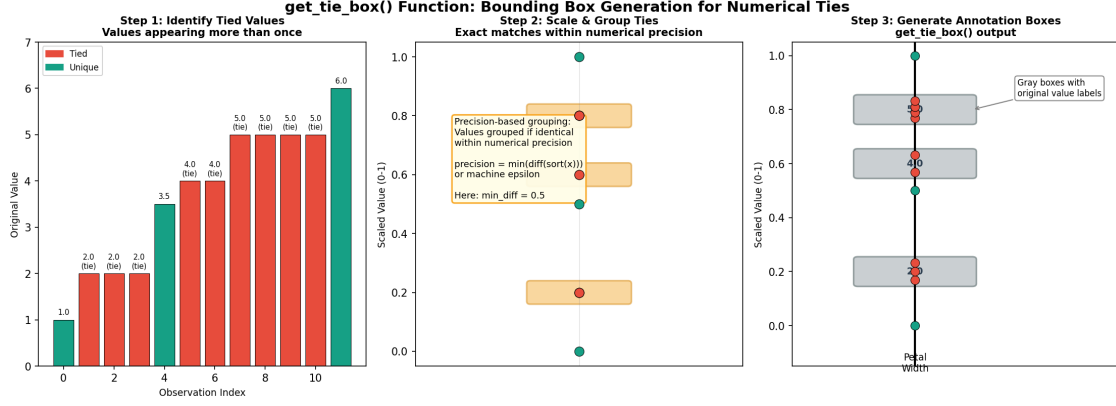


Figure 15: The tie box detection process. (1) Identify duplicated values, (2) scale and group contiguous spreads, (3) generate bounding box coordinates for annotation.

3.5.2 Tie Box Detection

The `get_tie_box()` function identifies and bounds tie regions for visualization:

3.6 Algorithm Summary

Algorithm: Unified Tie-Breaking

Input: Data D , variables $X_1 \dots X_p$, method $\in \{\text{from-left, from-right, from-both}\}$

Output: Adjusted positions $_{ij}$

1. Scale all variables to $[0, 1]$
2. For each axis j , compute resolution: $\delta_j = \min|y_1 - y_2|$ for distinct $y_1, y_2 \in Y_j$
3. Set processing order based on method:
 - from-left: process axes $1, 2, \dots, p$; reference left neighbor
 - from-right: process axes $p, p-1, \dots, 1$; reference right neighbor
 - from-both: process outward from center axis; reference previously processed neighbor
4. For each axis j in processing order:
 - a. Identify tied values: $V = \{v : |T_j(v)| > 1\}$
 - b. For each $v \in V$:
 - i. Set available space: $S(v) = \delta_j$
 - ii. Compute bounds: $L(v) = v - \frac{\delta_j}{2}, U(v) = v + \frac{\delta_j}{2}$
 - iii. Order $T_j(v)$ by $\kappa_i = \tilde{y}_{i,A_j}$ (positions on adjacent axis; right neighbor under **from-right**, left neighbor under **from-left**)
 - iv. Compute spacing: $\Delta = \frac{S(v)}{(k-1)}$
 - v. Assign positions: $_{i_m,j} = L(v) + (m-1) * \Delta$ for $m = 1, \dots, k$
5. Return adjusted data

3.7 Parameters

Table 3: Algorithm parameters

Parameter	Symbol	Default	Description
Band separation	σ	0.90	Fraction of δ_j used as band width; $(1 - \sigma) \cdot \delta_j$ is preserved as gap between adjacent bands
Boundary buffer	β	0.05	Fallback when no neighbor exists
Minimum spacing	δ	0.01	Ensures minimum separation
Tie tolerance	ε	10^{-10}	For floating-point comparison

4 Implementation and Evaluation Plan

4.1 Phase 1: Algorithm Development (Weeks 1-4)

Deliverable: Functional R package extension with comprehensive documentation Tasks:

1. Extend `pcp_arrange()` to detect and handle numerical ties
2. Implement `optimize_spacing_numerical()` or `tie_spacing()` function with perceptually-motivated parameters
3. Handle edge cases while preserving perceptual properties:
 - Single unique value (centered position)
 - Extreme skew (adaptive buffer sizing)
 - Missing values (explicit separation region)
4. Integration testing with existing ggpcp workflow

Code Structure (Simplified):

4.2 Phase 2: Perceptual Validation Studies (Weeks 5–7)

Success Criteria. Before detailing the study design, we specify the criteria that would constitute evidence of success for RQ2 and RQ3. For RQ2, we consider the algorithm successful if GPCP with tie-breaking produces an absolute percentage error on magnitude estimation tasks that is significantly lower (two-tailed $\alpha = 0.05$) than the hammock plot baseline, with $d \geq 0.4$. For ordinal and ratio tasks, we require accuracy rates at least 10 percentage points higher than baseline. For RQ3, we require that tracing accuracy in GPCP with tie-breaking exceeds that of the standard PCP without tie-breaking by at least 15 percentage points. Outcomes that do not meet these

thresholds will be interpreted as evidence that the algorithm achieves legibility parity but does not strictly surpass the comparison conditions.

Study 1: Frequency Perception

- Duration: Weeks 5–7
- Participants: $n = 128$ (64 per condition)
- Design: Between-subjects comparison of GPCP vs. hammock plot representations

Sample Size Justification:

A power analysis was conducted using G*Power 3.1 (Faul et al. 2007) to determine the minimum sample size required to detect a meaningful difference between visualization conditions. Based on prior graphical perception studies—including Heer and Bostock’s (2010) crowdsourced replication of Cleveland and McGill (1984), which used approximately 50 participants per condition, and Hofmann and Vendettuoli’s (2013) common angle plot study with 46 participants, we anticipated a medium effect size (Cohen’s $d = 0.5$). For a two-tailed independent samples t-test with $\alpha = 0.05$ and $power = 0.80$, the required sample size is $n = 64$ per group (total $N = 128$). This sample size also provides 80% power to detect effects as small as $d = 0.50$, which represents a practically meaningful difference in task accuracy between visualization types (Brysbaert 2019). To account for potential exclusions due to failed attention checks or incomplete responses (estimated at 10–15% based on crowdsourcing norms), we will recruit $n = 75$ per condition.

Procedure:

1. Magnitude estimation (12 trials): “What percentage have value X?”
2. Ordinal comparison (12 trials): “Which value is more frequent?”
3. Ratio judgment (12 trials): “A is how many times B?”
4. Confidence ratings (7-point scale) for each response

Analysis:

- Absolute percentage error for magnitude estimation
- Accuracy rates for ordinal and ratio tasks
- Bias analysis: systematic over/under-estimation
- Confidence calibration: accuracy vs. subjective confidence
- Effect sizes reported as Cohen’s d with 95% confidence intervals

4.3 Phase 3: Comparative Benchmarking (Weeks 6–10)

Visual Clutter and Information Density. Visual clutter constitutes a fundamental limitation on visualization effectiveness, occurring when too many visual elements compete for attention and overwhelm the viewer’s capacity to extract meaningful patterns. The two approaches manage clutter through fundamentally different mechanisms. For equispaced lines, clutter increases with the number of observations, but hierarchical sorting minimizes line crossings and thereby reduces visual complexity. The approach exhibits graceful degradation: as density increases, individual lines transition perceptually into ribbons and eventually into filled areas, yet individual observations remain theoretically accessible for highlighting or interactive selection. For constant-width

boxes, clutter depends primarily on the number of unique value combinations rather than on raw observation counts. Aggregation inherently reduces clutter, though at the cost of individual-level accessibility.

Dennig et al. (2021) formalize a suite of clutter metrics for Parallel Sets visualizations — including ribbon overlap, crossing angle, orthogonality, and ribbon width variance — and demonstrate that these metrics are mutually uncorrelated, measuring distinct properties of the display. These metrics can be adapted directly to compare the two approaches quantitatively here, and to guide the selection of axis and category orderings that minimize visual complexity for each dataset.

Computational Metrics:

1. Rendering performance (time, memory, scalability $n = 2$ to 100)
2. Visual quality metrics (Dennig et al. 2021): line crossing count, ribbon overlap, visual clutter index
3. Perceptual quality estimates: modeled eye movements, predicted visual search time

Case Studies:

- Palmer Penguins: Mixed categorical-numerical data with known correlation structure
- Iris Dataset: Multiple numerical variables with natural ties
- Asthma Data (Schonlau and Yang 2024): Direct comparison with published hammock plot

4.4 Phase 4: Integration and Dissemination (Weeks 8–12)

Deliverables:

1. R Package Update:
 - CRAN submission with full documentation
 - Vignette: “Handling Numerical Ties in Parallel Coordinate Plots”
 - Unit tests achieving $> 95\%$ coverage
2. Academic Paper:
 - Target: IEEE Transactions on Visualization and Computer Graphics
 - Submission deadline: May 2026 (IEEE VIS)
3. Supplementary Materials:
 - Open Science Framework repository with pre-registration
 - All experimental stimuli and data
 - Reproducibility package with power analysis scripts

5 Limitations and Future Directions

5.1 Study Limitations

- **Sample Characteristics:** University student population may not generalize to domain experts
- **Experimental Constraints:** Laboratory tasks may lack ecological validity
- **Scope:** Limited to static visualizations; interactive features not evaluated

5.2 Future Extensions

- Longitudinal study with domain experts in real analytical workflows
- Investigation of interaction effects between individual differences and visualization method
- Hybrid approaches: smooth interpolation between lines and boxes based on dataset properties
- Integration with animation and interactive highlighting techniques

6 Conclusion

Leveraging tie-breaking and sorting for numerical variables addresses a known issue with GPCPs while maximizing the numerical scale information available to the reader. The proposed equispaced line method extends ggpcp’s established categorical tie-breaking algorithm to numerical variables, creating a unified framework for handling ties across variable types. Implementing numerical tie-breaking preserves the Gestalt continuity of a single observation across parallel axes and allows the viewer to more easily recreate the joint density across multiple axes, relative to the standard numerical axis approach.

It is worth distinguishing between two uses of box-like visual elements in parallel coordinate displays. Hammock plots use parallelograms to represent aggregated bivariate relationships, where the area of each segment encodes the joint frequency of category pairs across adjacent axes. This approach scales well with large datasets but does not permit tracing of individual observations. GPCPs take a different approach: when multiple observations share the same value, their vertical positions are spread within a frame defined by the global resolution of the axis—the smallest difference between any two distinct values. This frame signals to the reader that positions within it reflect tie-breaking rather than meaningful data variation, while individual lines remain continuous and traceable.

Rather than viewing these as competing representations, they can be understood as complementary. The GPCP approach is optimal when the analytical task involves following specific cases or detecting outliers, whereas the hammock approach is optimal when the task involves comparing category frequencies or understanding distributional patterns without regard to individual observations. With many observations, equispaced lines may coalesce into ribbon-like bands; however, the principle of good continuation still applies, allowing viewers to perceive flow patterns even when individual lines are indistinguishable.

This research contributes to theory by demonstrating how Gestalt principles and models of preattentive processing generate testable hypotheses about visualization effectiveness. It contributes to practice by providing open-source software and evidence-based design guidelines. And it contributes to methodology by illustrating theory-driven, multi-method evaluation—computational

benchmarking to verify visual quality before perceptual testing, followed by appropriately powered user studies grounded in the graphical perception literature.

Effective visualization must be rooted in how people actually perceive: how they use principles of proximity, similarity, continuity, and common fate to organize and interpret visual features. This study takes that principle seriously by asking not merely “which visualization looks better?” but “which visualization aligns more effectively with perceptual organization principles for specific analytical tasks?” The answer, as with most questions in visualization, will depend on context. But it will now rest on empirical evidence and perceptual theory rather than assumption.

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